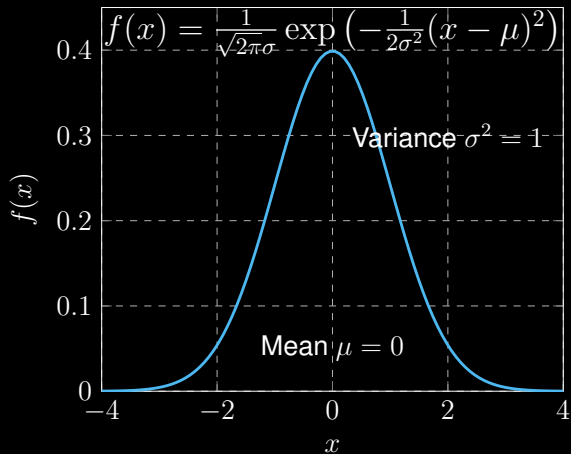


Standard Gaussian Distribution



Multivariate Gaussian Distribution

Probability density function:

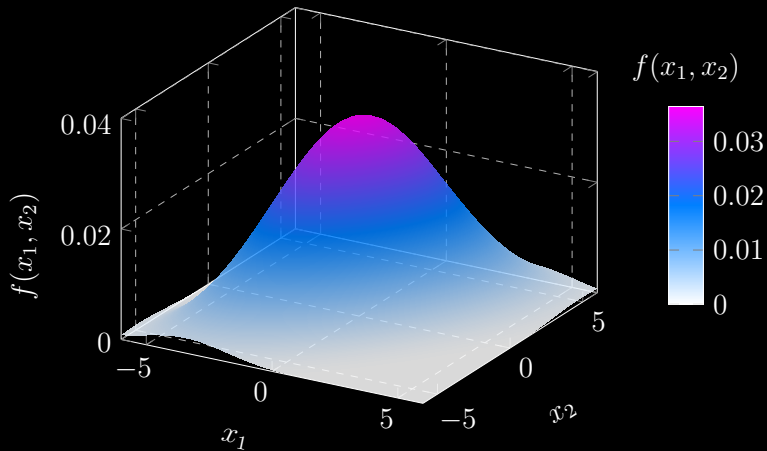
$$f(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} \sqrt{\det(\mathbf{\Sigma})}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

with mean vector $\boldsymbol{\mu} \in \mathbb{R}^n$ and covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{n \times n}$.

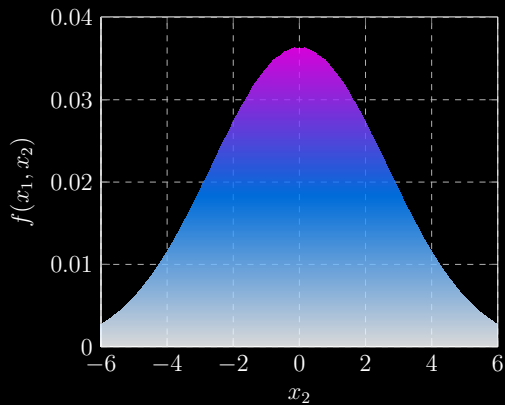
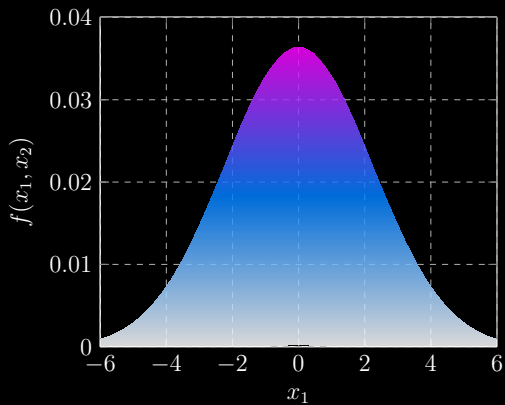
Bivariate Gaussian distribution:

$$\text{Mean vector } \mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and covariance matrix } \Sigma = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$$

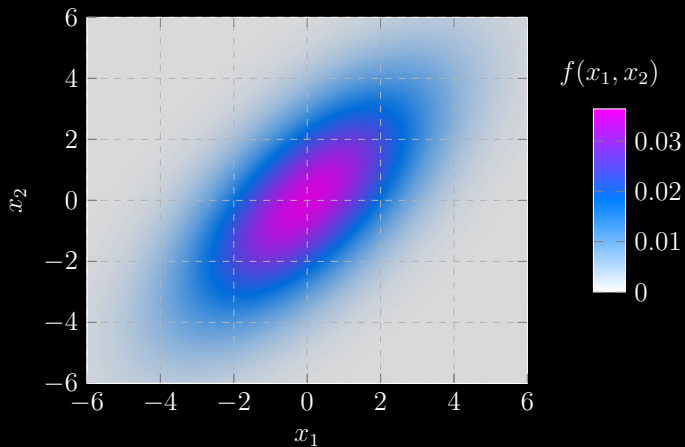
Bivariate Gaussian Distribution



Bivariate Gaussian Distribution



Bivariate Gaussian Distribution



The **level set** of a Gaussian distribution is the **set of points** in n -dimensional space with the **same probability density**.

$$\frac{1}{(2\pi)^{n/2} \sqrt{\det(\Sigma)}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right) = c_0 \quad (\text{at level } c_0)$$

$$\Rightarrow \quad (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c \quad (\text{with constant } c)$$

The **level set** of multivariate Gaussian distribution is **an ellipsoid** in the n -dimensional space, why?

$$\begin{aligned} & (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^\top \right)^{-1} (\mathbf{x} - \boldsymbol{\mu}) \end{aligned}$$

The **level set** of multivariate Gaussian distribution is **an ellipsoid** in the n -dimensional space, why?

$$\begin{aligned} & (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^\top \right)^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\sum_{i=1}^n \frac{1}{\lambda_i} \mathbf{u}_i \mathbf{u}_i^\top \right) (\mathbf{x} - \boldsymbol{\mu}) = c \end{aligned}$$

The **level set** of multivariate Gaussian distribution is **an ellipsoid** in the n -dimensional space, why?

$$\begin{aligned} & (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^\top \right)^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\sum_{i=1}^n \frac{1}{\lambda_i} \mathbf{u}_i \mathbf{u}_i^\top \right) (\mathbf{x} - \boldsymbol{\mu}) = c \\ \Rightarrow & \frac{(\mathbf{u}_1^\top (\mathbf{x} - \boldsymbol{\mu}))^2}{\lambda_1} + \dots + \frac{(\mathbf{u}_n^\top (\mathbf{x} - \boldsymbol{\mu}))^2}{\lambda_n} = c \quad (\text{ellipsoid}) \end{aligned}$$

with **eigenvalue** λ_i and **eigenvector** $\mathbf{u}_i$.

Bivariate Gaussian distribution:

$$\text{Mean vector } \boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and covariance matrix } \boldsymbol{\Sigma} = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$$

The ellipse:

$$\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \frac{(\mathbf{u}_2^\top \mathbf{x})^2}{\lambda_2} = c$$

with eigenvalues λ_1, λ_2 and eigenvectors $\mathbf{u}_1, \mathbf{u}_2$.

Covariance Matrix $\Sigma = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$

Eigenvalue decomposition:

$$\Sigma = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 1.877 \quad \lambda_2 = 10.123$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} -0.788 \\ 0.615 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.615 \\ -0.788 \end{bmatrix}$$

Covariance Matrix $\Sigma = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$

Eigenvalue decomposition:

$$\Sigma = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

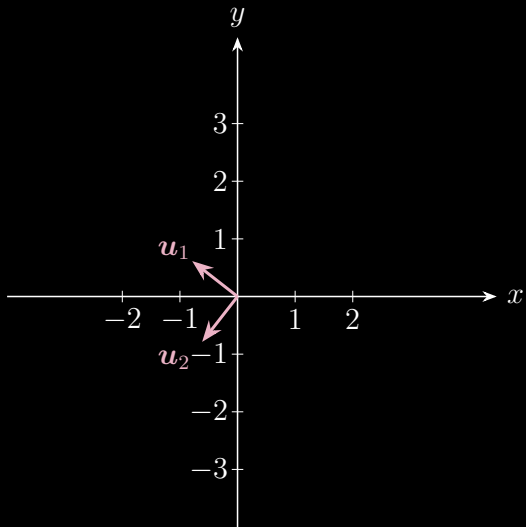
$$\lambda_1 = 1.877 \quad \lambda_2 = 10.123$$

and eigenvectors:

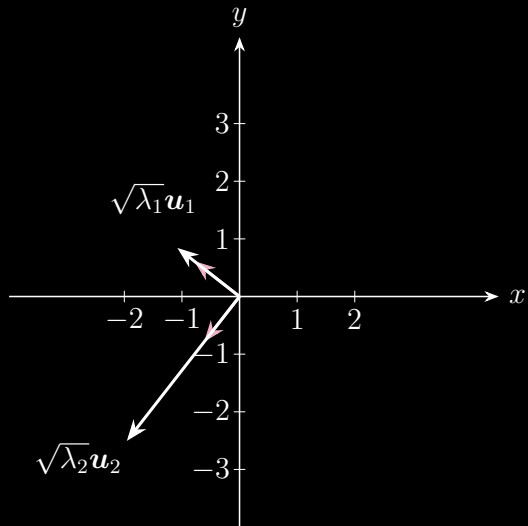
$$\mathbf{u}_1 = \begin{bmatrix} -0.788 \\ 0.615 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.615 \\ -0.788 \end{bmatrix}$$

```
1 import numpy as np
2
3 S = np.array([[5, 4], [4, 7]])
4 eig_val, eig_vec = np.linalg.eig(S)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```

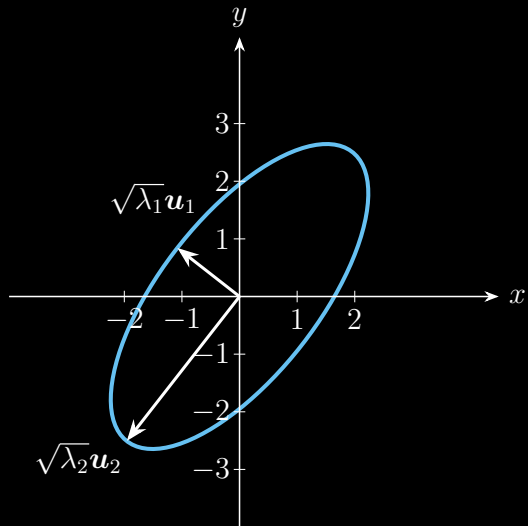
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{(\sqrt{\lambda_1})^2} + \frac{(\mathbf{u}_2^\top \mathbf{x})^2}{(\sqrt{\lambda_2})^2} = 1$



Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{(\sqrt{\lambda_1})^2} + \frac{(\mathbf{u}_2^\top \mathbf{x})^2}{(\sqrt{\lambda_2})^2} = 1$

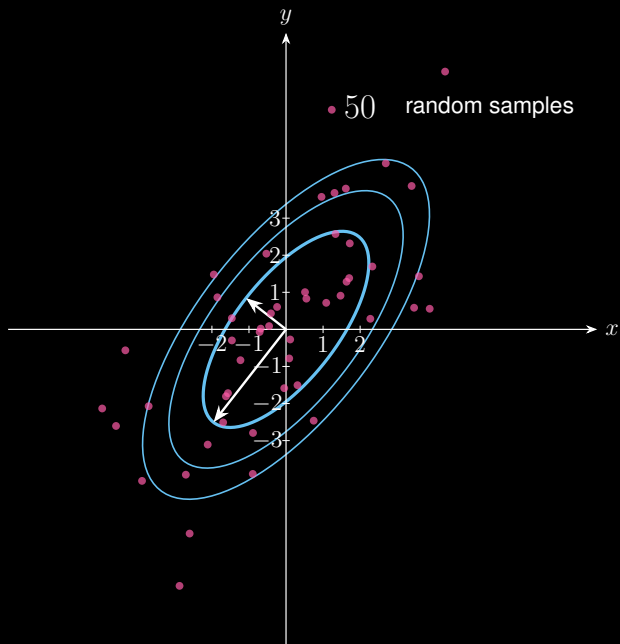


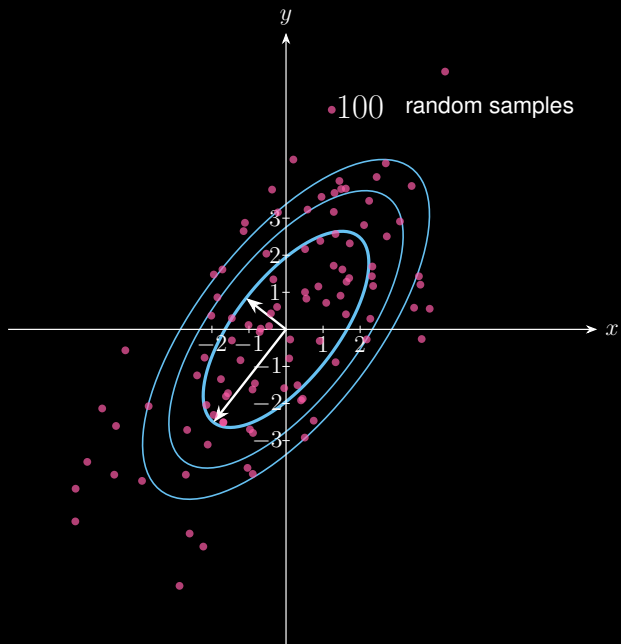
Ellipse $\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{(\sqrt{\lambda_1})^2} + \frac{(\mathbf{u}_2^\top \mathbf{x})^2}{(\sqrt{\lambda_2})^2} = 1$

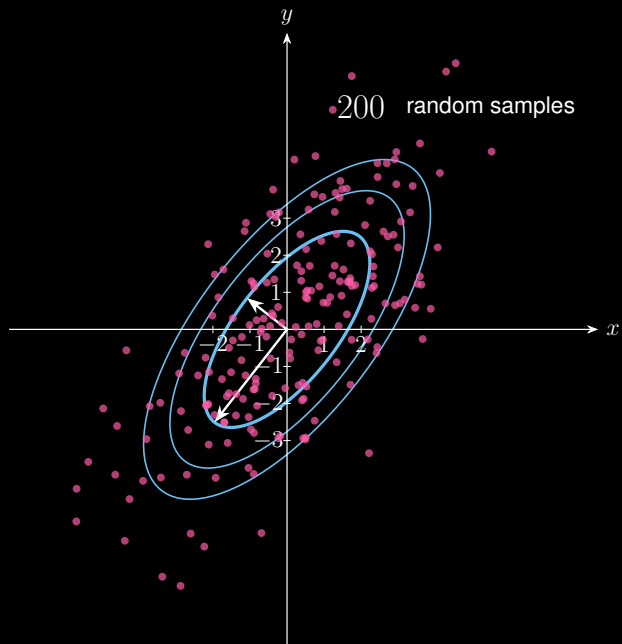


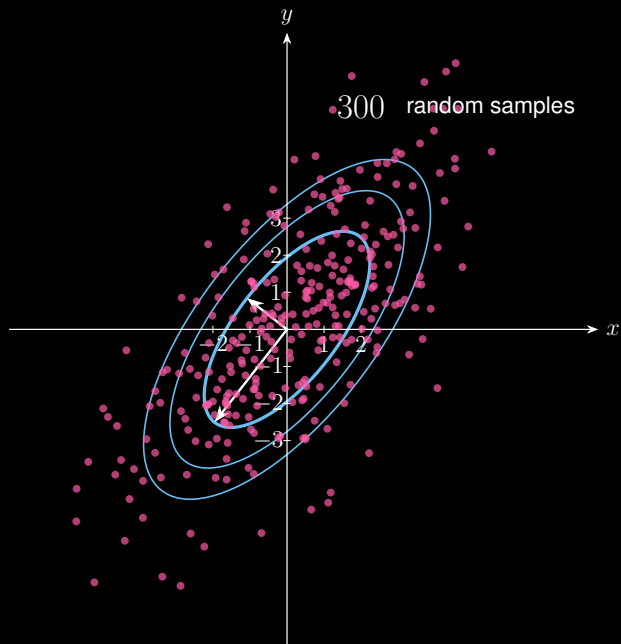
Generate random samples with $\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$

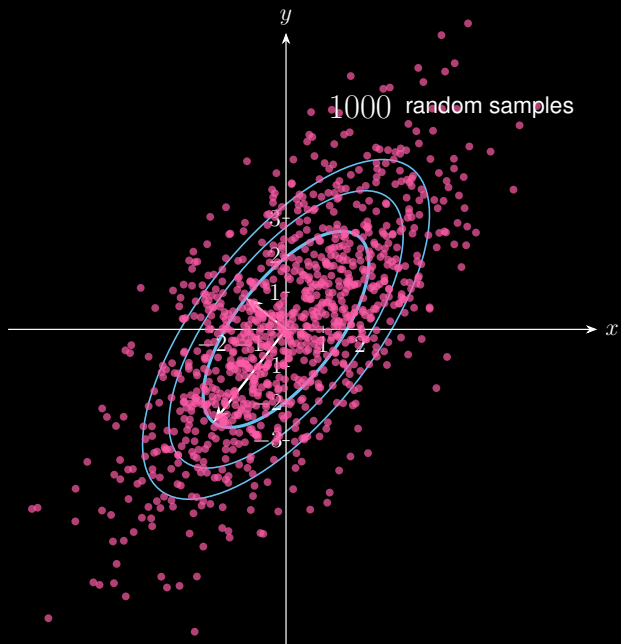
```
1 import numpy as np
2 np.random.seed(0)
3
4 n = 1000
5 mu = np.zeros(2)
6 Sigma = np.array([[5, 4], [4, 7]])
7 mat = np.random.multivariate_normal(mu, Sigma, n)
8 np.savetxt('sample{}.txt'.format(n), mat)
```





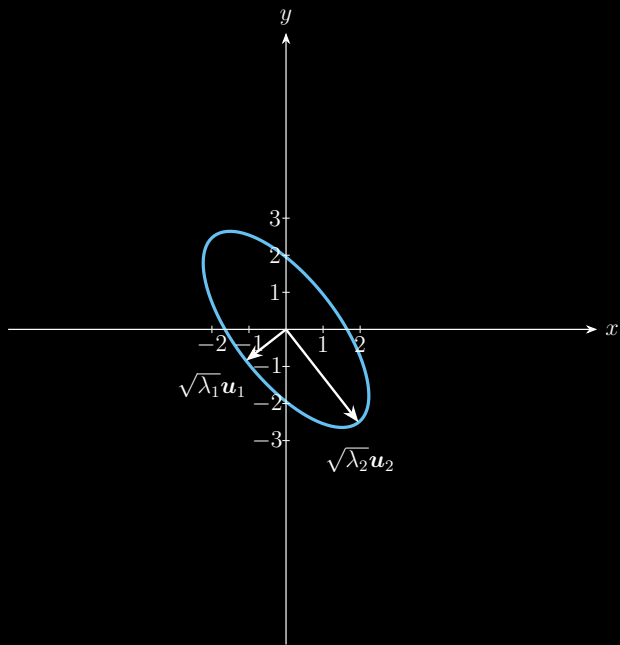


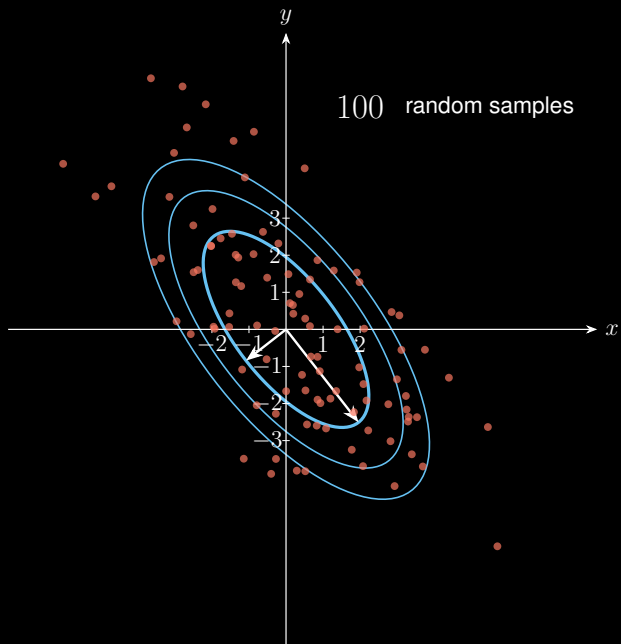


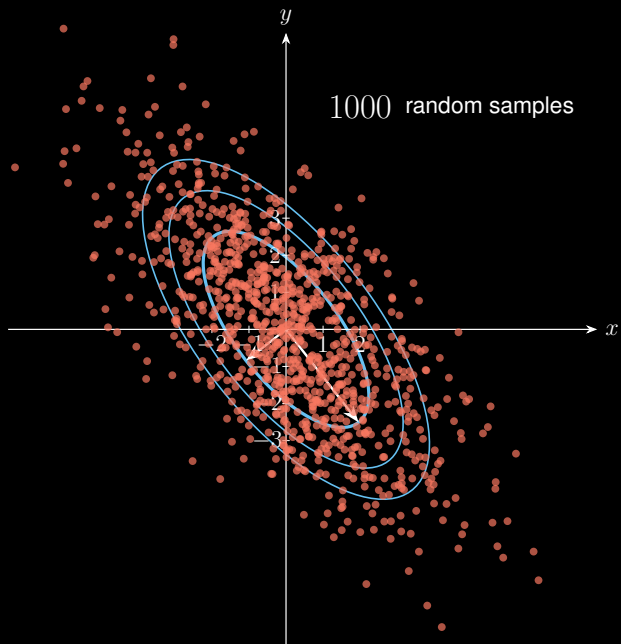


Bivariate Gaussian distribution:

$$\text{Mean vector } \boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and covariance matrix } \boldsymbol{\Sigma} = \begin{bmatrix} 5 & -4 \\ -4 & 7 \end{bmatrix}$$

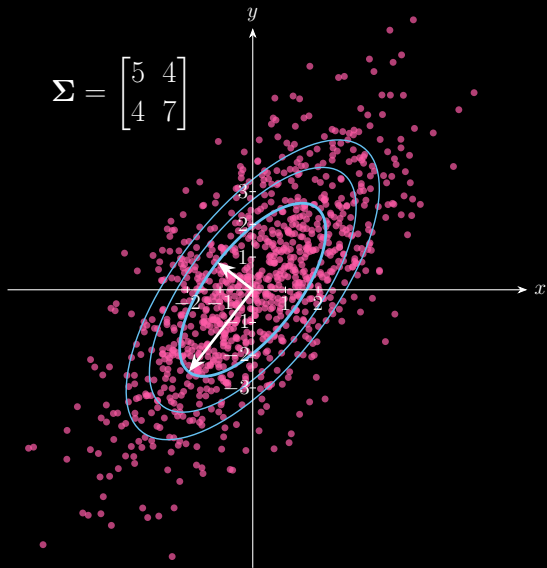




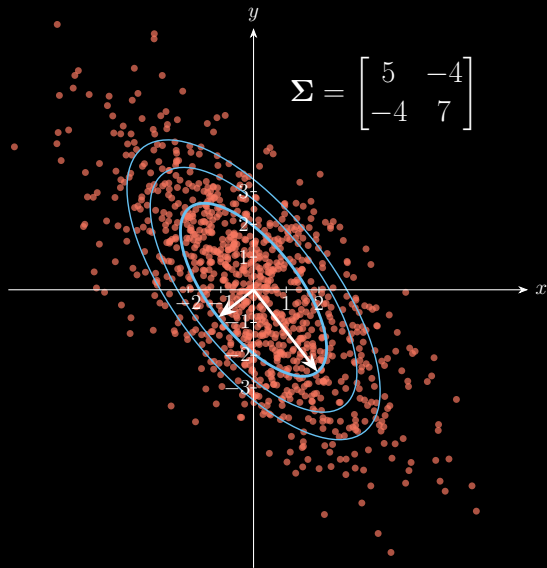
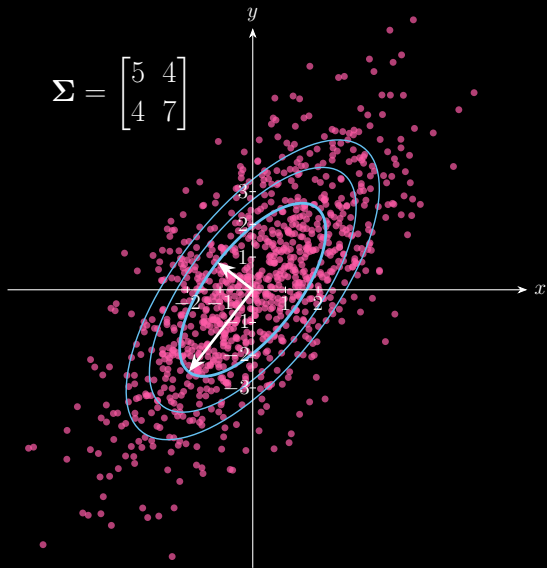


Eigenvalue Decomposition Defines the Shape of Multivariate Gaussian

$$\Sigma = \begin{bmatrix} 5 & 4 \\ 4 & 7 \end{bmatrix}$$



Eigenvalue Decomposition Defines the Shape of Multivariate Gaussian



Thanks for your attention!

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 How to reach me: chenxy346@gmail.com